



Database Modeling Database Normalization for the Project

Aixa Elenka Mendoza Filisola | A01782727

Oscar Miguel Lara Elizondo | A01781855

José Emilio Lara Posada | A01782838

@2026 ITARI Games Studio

Todos los derechos reservados

Tables

No improvements implemented by using a generative model

The database schema was revised to improve normalization and avoid redundancy.

1. The intermediary table "Card_Effects" was created to remove multi-valued attributes and allow a single card to contain multiple effects. This table serves as a dedicated table to the attribute "effect" that existed in the original "Card" table.
2. The attribute "Time_stats" in the "Player" table was expanded to avoid storing multiple statistics in one field. In the final version it is divided into the specific statistics it considered: "average_race_time", "fastest_lap", "total_play_time"- This allows queries with this data to become easier to implement.
3. In the "player_game" table we expanded the attribute "final_score" into "total_play_time" and "fastest_lap" to avoid multiple statistics.
4. Implemented attributes in the "Users" table that maintain the account information and the gameplay information separate. These include the "email", "username", "password", "age". All of this based on the third normal form.
5. We focused the "CARD_STATS" table to tracking data from the "CARD" entity and eliminated an attribute to avoid unnecessary null values.
6. For the "Rivals" table, formerly known as enemies, we added a field to count the times a rival may appear during the match/race. This creates a relationship between the table "GAMESESSION" of 1:N. This way we avoid unnecessary duplication and repeating rival information inside game records.

Importance of Normalization

Normalization is an important process to design and implement a database. These practices reduce redundancy, inconsistency, maintenance issues and organize data efficiently. By implementing the normalization principles diagrams become more scalable and better structured leading to better data bases.

During the development process, generative language tools contributed as supportive tools or assistants during the design process by suggesting improvements in the relationships between entities and providing answers to questions about keys and intermediary tables. Considering alternative viewpoints can be useful to identify structural issues that might otherwise be overlooked.

However, the development work itself remains the responsibility of the designers and programmers. The developers understand the game mechanics, game function, entities and relationships that are required for the project. Final design choices and adjustments have to remain in the hands of the programmers because the creativity, functionality and implementation goals originate from human decision-making. In other words the generative language tools serve as support tools not the actual designers of the system. The final approval, validation and implementation of the database structure remains the responsibility of the development team.